

Lithium-ion Batteries

Portable Power Source For Portable Devices

Portable devices must have a power source as portable as possible. With ample energy density in a smaller size and lesser weight, lithium-ion batteries have made inroads in every portable device



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Have you ever been in a situation where you had to make an urgent call and your cell-phone ran out of power, or your laptop switched off in the middle of an important client presentation? Clearly, using gadgets on the move is not fashion but a necessity to stay ahead in the competition. The gadgets need a portable power source to work unplugged. This has created a huge platform for R&D in batteries—the portable power storage devices.

When it comes to deciding a bat-

tery for your new product, it is very important to look at the battery chemistry. Hiren Shah, assistant general manager, Panasonic Industrial India, shares, "A battery is an electricity generator whose performance depends on its chemical reaction." The battery has a shelf life and service life, and its performance varies with changes in environmental conditions. Therefore before finalising the battery for your portable application, its complete know-how is extremely important.

Today, about 90 per cent of the consumer products use a Lithium-ion (Li-ion) battery. These hold a market

value of \$11 billion (as of 2010), which is expected to reach \$43 billion by 2020.

Syed Mubasheer Ali, business development manager, D Espot, shares, "Li-ion batteries are common in small applications like micro-robots, mobile phones, field-deployable devices, cameras and medical devices. With the kind of benefits these offer, they have also entered and will soon establish a market for electric and hybrid vehicles."

Expectations from a battery

Packing more power in a small area is a key requirement in every consumer product today. As laptops, tablets and cellphones shrink in size and weight without compromising on the quality of audio, graphics, etc, you need batteries which are thinner and lighter with higher power densities.

The power and voltage requirements vary from product to product, and commercially available batteries don't always meet that level of requirement. S.S. Kandhari, CEO, Kandhari Photoelectronics, says, "These needs are served by supplying battery packs ranging from 3.7V to 14.8V. Li-ion battery packs are much lighter and smaller than other battery chemistries, making them highly suitable for LED lighting and UPS backup applications."

Ali shares, "Starting from voltage and playtime requirements to the preference of dimensions and weight, you need to make mathematical estimates